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Jump-starting verification via direct outreach

Executive Summary

Compute verification is probably the most urgent outstanding problem in AI safety today. Enabling it would allow rivals to pace or even pause the frontier, secure in the knowledge that other participants aren't sneaking ahead. In spite of that, it's critically labor-constrained: estimates are that only a few dozen people are working on it globally; for FTE it's around 25.

My theory of change is especially crisp: give academic researchers money in order to have them work on specific problems for which they are already one of the world's top experts. This dates back at least to Vannevar Bush, who jump-started the Manhattan Project by working directly with a handful of universities. The track record shows it can be incredibly fast: it has the potential to double or triple the amount of labor working on verification **before New Year's Eve**.

Matching problems to experts

Unlike many problems in AI Safety, verification is not dual-use. There's a well-defined set of subproblems that need to get solved. These problems also don't require upskilling new entrants from scratch. On the contrary, most of these subproblems are active areas of research for various academics; it's just that those researchers are largely unaware of AI safety and the potential benefit of their work. Rather than diverting the scarce pool of AI Safety researchers onto verification, I propose **growing the community** by reaching out to the academics directly, proactively offering funding. If, over the course of the verification work, some are especially value-aligned and choose to enter the field, so much the better.

Money moves research fast

We know this approach works: we saw it recently when "Fast Grants" were assigned during COVID. In April 2020, Tyler Cowen and team set up a fund with a 30-minute application and a promise of decisions within 48 hours; over the course of 2020 it moved more than $50 million to over 260 researchers. In surveys, 64% of respondents said that the research wouldn't have happened at all without the FastGrant. So well-directed money can change the direction of research on very short timescales, and researchers I've spoken with confirm that they have done so and would again.

Who, and how many

What scale are we discussing? I've reviewed 52 documents that lay out the open problems (from RAND, GovAI, IAPS and others, plus the chip-security and IAEA safeguards literature), and identified less than 200 specific subquestions that need to be solved, and over 600 specialist academics who are actively publishing on precisely those questions; full list at:
canaryinstitute.ai/research/open-problems

The mechanism is fast and personal

Most universities have a "gift" mechanism by which research can be directed to a particular professor, and even to a particular project, with minimal overhead (often less than 10%). There are often some minor stipulations, but public-benefit work funded by a nonprofit should clear them handily. Donors typically provide a letter which specifies the faculty member and topic, but can't specify deliverables or dates. In this scenario, those are carried by the personal relationship between the funder and the specialist.

And in this proposal there *would* be a personal relationship: these requests wouldn't be coming from cold-calls to the specialist faculty. I've mapped which coauthors the specialists share with people already working on verification. These "seed" researchers could contact the mutual coauthors and ask for a warm introduction, then explain the problem and the AI risk scenario, and offer the funding. I estimate over 300 academic specialists can be reached in this manner. 44 of our specialists are located at just four universities (Stanford, Georgia Tech, U of Florida, and U of Maryland), and 33 of those are already reachable.

The pilot: one university, ten researchers, $1.2M

I'm currently contacting likely researchers at those four universities to see how they respond. The pilot would go to whichever one is most receptive: roughly ten specialist faculty at that school, $120k each, for $1.2M in total. That's enough for each to fully fund a graduate student for a year to work on the problem under the supervision of one of the world's leading experts. Keeping the pilot on one campus means one gift office and one set of stipulations, and the researchers can talk to each other down the hall. Since money directed in this manner arrives quickly, the response time would be comparably short.

Why move now

The pilot can move this fast because it uses students who are already enrolled: a professor can move an existing student onto a new project within weeks. For longer-term commitment, funders may need to act soon. Departments decide graduate admissions over the winter, mostly January through March, and they admit only as many students as they have money to support. A professor who has a gift letter by then can admit an extra student for the fall, whereas one who gets it in April has to wait a year.